

Week 3 Boundary Detection

Week 3 Quiz

Graded Assignment: Week 3 Boundary Detection

Your grade: 100%

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Week 3 Boundary Detection

1. Given the points (1, 1), (2, 1), (2, 1), (4, 3) and (5, 3), find the slope m and y -intercept c of the line of best fit using the vertical distance least squares method. The estimated line parameters are:

☐ $m = 5, c = 1$

☐ $m = 6, c = 0$

☐ $m = 0.1, c = 0.7$

☒ $m = 0.6, c = 0.2$

Correct

Due Oct 19, 7:32 PM EDT

Attempts 1/3

Use the closed form solution

Try again

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2. For the line formulation $E = \frac{1}{2} \sum_{i=1}^n (x_i \sin \theta - y_i \cos \theta + \rho)^2$, which is the correct equation for $\frac{\partial E}{\partial \theta}$?

☐ $\frac{1}{2} \sum_{i=1}^n (x_i \sin \theta - y_i \cos \theta + \rho)$

☐ $\frac{1}{2} \sum_{i=1}^n (x_i \sin \theta - y_i \cos \theta + \rho)^2$

☒ $\frac{1}{2} \sum_{i=1}^n (x_i \sin \theta - y_i \cos \theta + \rho)$

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Correct

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3. Suppose we want to fit a curve to our edge points with the equation $y = mx^2 + nx^2 + p$. Which of the following is the objective function we want to minimize?

☐ $E = \frac{1}{2} \sum_{i=1}^n (y_i - mx_i^2 - nx_i^2 - p)^2$

☐ $E = \frac{1}{2} \sum_{i=1}^n (y_i + mx_i^2 + nx_i^2 + p)^2$

☒ $E = \frac{1}{2} \sum_{i=1}^n (y_i - mx_i^2 - nx_i^2 - p)^2$

☐ $E = \frac{1}{2} \sum_{i=1}^n (y_i - mx_i^2 - nx_i^2 - p)^2$

Correct

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4. Given the coordinates of several points on a polynomial, the parameters of the polynomial can be found using:

☐ Non-linear system of equations

☒ Linear system of equations

☐ Differential equation

☐ Quadratic equation

Correct

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5. We want to fit a line of the form $y = ax^2 + bx + c$ to the points (1,1), (2,2) and (3, 1). Using a linear system, what are the best-fit values of a , b and c ?

☐ $a = 3, b = -2, c = -2$

☐ $a = -1, b = -4, c = 5$

☒ $a = -1, b = 4, c = -2$

☐ $a = 1.5, b = 1, c = 0$

Correct

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6. What is the pseudo-inverse of the following matrix: $\begin{bmatrix} 2 & 4 \\ 0 & 3 \\ 0 & 1 \end{bmatrix}$

☐ $\begin{bmatrix} 1/10 & 4/10 \\ 0 & 2/5 \end{bmatrix}$

☐ $\begin{bmatrix} 1/10 & 4/10 \\ 0 & 2/5 \end{bmatrix}$

☒ $\begin{bmatrix} -3/35 & 19/210 & 1/14 \\ 2/7 & -1/42 & -1/14 \end{bmatrix}$

☐ $\begin{bmatrix} -6 & 6 & 1 \\ 20 & -1 & 5 \end{bmatrix}$

Correct

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7. Given a continuous curve $\mathbf{V}(x) = (x(x), y(x))$, it can be made smooth by minimizing:

☐ $\int \|\mathbf{V}\|^2$

☐ $\int \|\mathbf{V}\|^2$

☐ $\int \|\mathbf{V}\|^2$

☒ $\int \|\mathbf{V}\|^2$

Correct

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8. Given a point (x_i, y_i) on a continuous curve $\mathbf{V}$, how many neighboring points (in addition to the point itself) do you need to approximate the elasticity of the curve at (x_i, y_i) using finite differences?

☐ 1

☐ 2

☐ 3

☐ 4

Correct

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9. Which of the following statements about accumulator arrays for line detection is false?

☐ Voting over a patch of cells makes up for inaccurate edge detection.

☒ A small cell size may reduce the effect of noise but causes a computational blowup.

☐ A large cell size may merge close but distinct lines.

☐ In a perfect accumulator array, the number of lines is equivalent to the number of peaks in the array.

Correct

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10. What defines the direction of an edge d_i at point i in the generalized Hough transform?

☐ Angle formed by line to the reference point

☐ Both the angle formed by line to the reference point and the distance to it

☐ Tangent to the edge at point i

☒ Normal to the edge at point i

Correct

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https://www.coursera.org/learn/features-and-boundaries/assignment-submission/24ghn/week-3-boundary-detection/view-feedback

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